MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 10: Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. A quadrilateral $ABCD$ circumscribes a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, then the length of side AD is: [1]

(A) 3 cm (B) 4 cm (C) 5 cm (D) 6 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 3 cm

Justification: For a circumscribed quadrilateral, $AB + CD = BC + AD \implies 6 + 4 = 7 + AD \implies AD = 10 - 7 = \boxed{3 \text{ cm}}$.

Q2. The length of the tangent drawn from a point A at a distance of 10 cm from the centre of a circle of radius 6 cm is: [1]

(A) 7 cm (B) 8 cm (C) $4\sqrt{2}$ cm (D) $2\sqrt{34}$ cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 8 cm

Justification: Let tangent point be T. In right $\triangle OTA$, $AT = \sqrt{OA^2 - OT^2} = \sqrt{10^2 - 6^2} = \sqrt{64} = \boxed{8 \text{ cm}}$.

Q3. If a chord AB subtends an angle of $60°$ at the centre of a circle, then the angle between the tangents at A and B is: [1]

(A) $60°$ (B) $90°$ (C) $120°$ (D) $150°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $120°$

Justification: Angle between tangents $= 180° - \angle AOB = 180° - 60° = \boxed{120°}$.

Q4. At one end of a diameter AB of a circle of radius 5 cm, tangent XAY is drawn. The length of chord CD parallel to XY at distance 8 cm from A is: [1]

(A) 4 cm (B) 6 cm (C) 8 cm (D) 5 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 8 cm

Justification: Since $OA = 5$ cm and $OA \perp XY$, chord $CD \parallel XY$ at 8 cm from A meets AB perpendicularly at E where $OE = 8 - 5 = 3$ cm. Half-chord $CE = \sqrt{5^2 - 3^2} = 4$ cm $\implies CD = 2CE = \boxed{\mathbf{8}\text{ cm}}$.

Q5. **Assertion (A):** If a chord of the larger of two concentric circles touches the smaller circle, then it is bisected at the point of contact.
Reason (R): The perpendicular drawn from the centre of a circle to a chord bisects the chord. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Radius of the smaller circle drawn to the point of contact is perpendicular to the tangent (which is a chord of the larger circle). By Reason (R), this perpendicular bisects the chord. Both (A) and (R) are true and (R) correctly explains (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

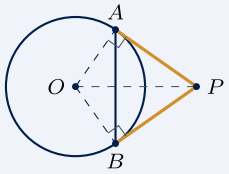
Q6. In a circle with centre O, tangents PA and PB are drawn from an external point P. If $\angle APB = 70°$, find the measure of $\angle OAB$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Supplementary Angle Property [1 Mark]
Since $\angle OAP = \angle OBP = 90°$, in quadrilateral $OAPB$:
$\angle AOB = 180° - \angle APB = 180° - 70° = 110°$.

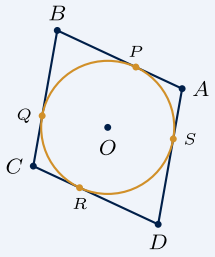
Step 2: Angle Sum in Isosceles $\triangle OAB$ [1 Mark]
Since $OA = OB$ (radii), $\angle OAB = \angle OBA$. Thus:

$2\angle OAB + 110° = 180° \implies 2\angle OAB = 70° \implies \boxed{\angle OAB = \mathbf{35°}}$



Q7. Prove that the parallelogram circumscribing a circle is a rhombus. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



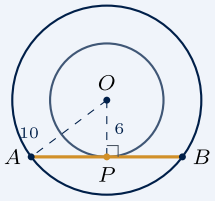
Given: Parallelogram $ABCD$ circumscribes a circle touching AB, BC, CD, DA at P, Q, R, S.
To Prove: $ABCD$ is a rhombus ($AB = BC = CD = DA$).
[0.5 Mark]

Proof & Deduction: [2.5 Marks]
1. Since tangents from an external point are equal (Theorem 10.2):
$AP = AS, \quad BP = BQ, \quad CR = CQ, \quad DR = DS.$
2. Adding all four equations: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \implies AB + CD = AD + BC.$
3. Since $ABCD$ is a parallelogram, $AB = CD$ and $AD = BC$.
Substituting gives $2AB = 2BC \implies AB = BC.$

$\boxed{\therefore AB = BC = CD = DA \implies ABCD \text{ is a rhombus} \quad \text{[Proved]}}$

Q8. In two concentric circles, a chord AB of the larger circle touches the smaller circle at P. Prove that P bisects AB. If radii are 10 cm and 6 cm, find the length of chord AB. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Let O be the common centre with $OA = 10$ cm and $OP = 6$ cm.

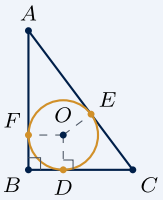
Step 1: Proving P Bisects AB [1.5 Marks]
AB is tangent to the smaller circle at P, so $OP \perp AB$ (Theorem 10.1). Since the perpendicular from centre O to chord AB of the outer circle bisects it, **AP = PB** (P bisects AB).

Step 2: Finding Length AB [1.5 Marks]
In right $\triangle OPA$, $AP = \sqrt{OA^2 - OP^2} = \sqrt{10^2 - 6^2} = \sqrt{64} = 8$ cm.

$\boxed{\text{Length of chord } AB = 2AP = \mathbf{16} \text{ cm}}$

Q9. A right triangle ABC right-angled at B has $AB = 8$ cm and $BC = 6$ cm. A circle with centre O is inscribed in $\triangle ABC$. Find the inradius r and the area of the incircle. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Hypotenuse Calculation [0.5 Mark]
In right $\triangle ABC$, $AC = \sqrt{AB^2 + BC^2} = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$ cm.

Step 2: Finding Inradius r [1.5 Marks]
Using tangents from vertices (where $BD = BF = r$ form square $ODBF$):
$r = \frac{AB + BC - AC}{2} = \frac{8 + 6 - 10}{2} = \frac{4}{2} = \mathbf{2}$ cm.
(Alternatively, Area $= r \cdot s \implies 24 = r \cdot 12 \implies r = 2$ cm.)

Step 3: Area of Incircle [1 Mark]
Area $= \pi r^2 = \pi(2)^2 = \mathbf{4\pi}$ cm^2 $\approx$ **12.57** cm^2 ($\frac{88}{7}$ cm^2 for $\pi = \frac{22}{7}$; or 12.56 cm^2 for $\pi = 3.14$).

$\boxed{r = \mathbf{2} \text{ cm}, \quad \text{Area of Incircle} = \mathbf{4\pi} \text{ cm}^2 \approx \mathbf{12.57} \text{ cm}^2}$

Q10. Bicycle Sprocket and Chain Drive Geometry

A bicycle transmission system features a circular chain sprocket of radius 6 cm with centre O. The upper and lower chain runs act as tangents TA and TB from an external derailleur guide T. The distance from the sprocket centre to the guide is $OT = 10$ cm.

(i) Determine the length of the tangent chain segment TA. [1]

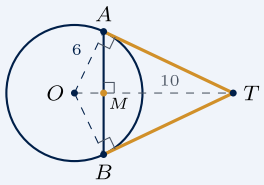
(ii) Find the value of $\sin(\angle OTA)$ in the right $\triangle OAT$. [1]

(iii) Calculate the length of the chord of contact AB between the tangent points, OR determine the perimeter of quadrilateral $OATB$. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Length of Tangent TA: [1 Mark]

Since $OA \perp TA$, in right $\triangle OAT$: $TA = \sqrt{OT^2 - OA^2} = \sqrt{10^2 - 6^2} = \boxed{8 \text{ cm}}$.



(ii) Value of $\sin(\angle OTA)$: [1 Mark]

In right $\triangle OAT$, $\sin(\angle OTA) = \frac{\text{Opposite } (OA)}{\text{Hypotenuse } (OT)} = \frac{6}{10} = \boxed{\frac{3}{5} = 0.6}$.

(iii) Main Choice — Length of Chord of Contact AB: [2 Marks]

Let OT intersect chord AB perpendicularly at midpoint M. In right $\triangle OAT$, Area$(\triangle OAT) = \frac{1}{2}(OA)(TA) = \frac{1}{2}(OT)(AM) \implies 6 \times 8 = 10 \times AM \implies AM = 4.8$ cm.

Therefore, chord $AB = 2 \times AM = \boxed{\mathbf{9.6} \text{ cm}}$.

— OR — (Alternative Choice — Perimeter of Quadrilateral $OATB$): [2 Marks]

Since $TA = TB = 8$ cm and $OA = OB = 6$ cm,

$$\boxed{\text{Perimeter}(OATB) = OA + TA + TB + OB = 6 + 8 + 8 + 6 = \mathbf{28} \text{ cm}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★